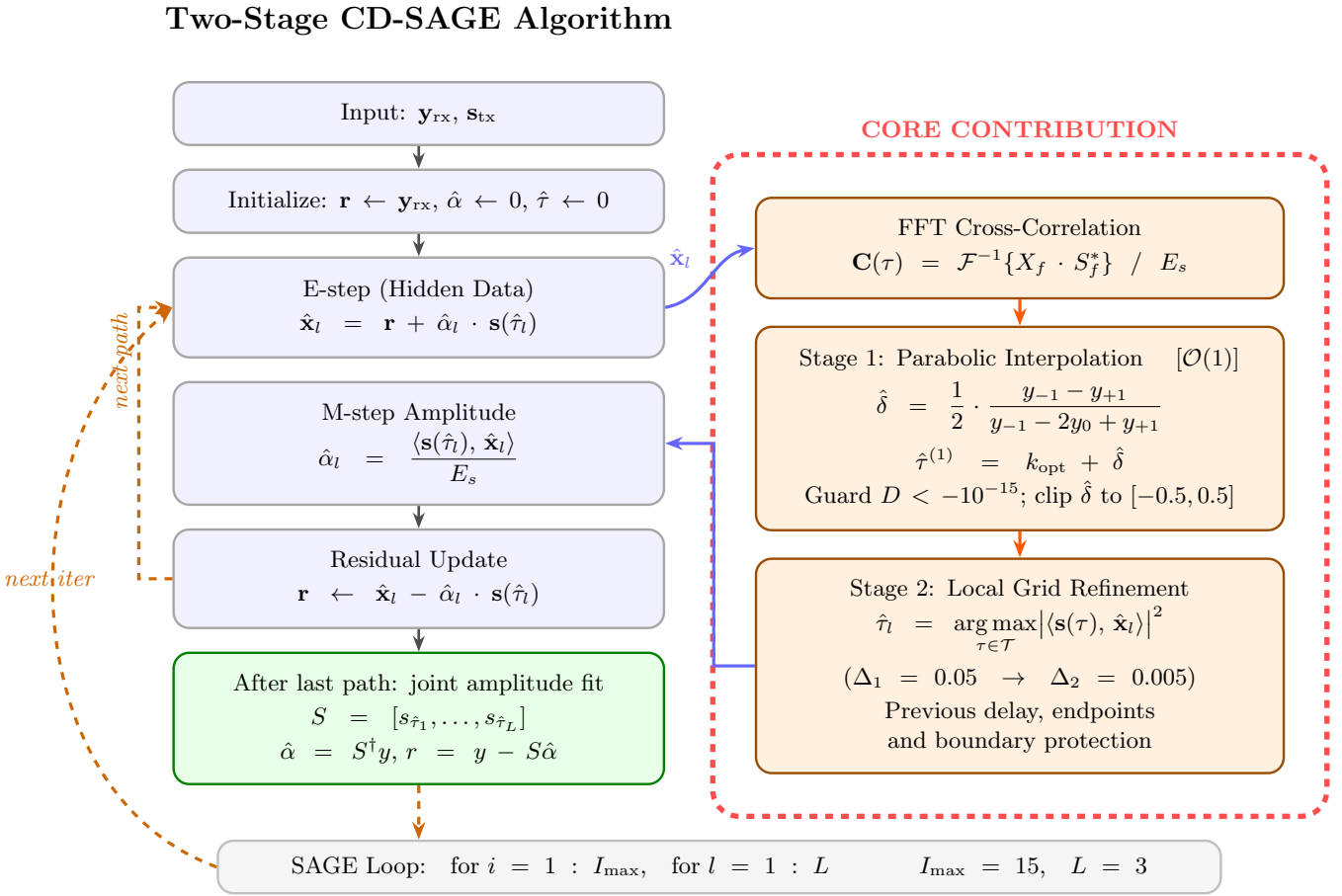

Two-Stage CD-SAGE Algorithm
Input: $\mathbf{y}_{\mathrm{rx}}$, $\mathbf{s}_{\mathrm{tx}}$
Initialize: $\mathbf{r} \leftarrow \mathbf{y}_{\mathrm{rx}}$, $\hat{\alpha} \leftarrow 0$, $\hat{\tau} \leftarrow 0$
E-step (Hidden Data)
$\hat{\mathbf{x}}_l = \mathbf{r} + \hat{\alpha}_l \cdot \mathbf{s}(\hat{\tau}_l)$
$\hat{\mathbf{x}}_l$
CORE CONTRIBUTION
FFT Cross-Correlation
$\mathbf{C}(\tau) = \mathcal{F}^{-1}\{X_f \cdot S_f^*\} \;/\; E_s$
Stage 1: Parabolic Interpolation $[\mathcal{O}(1)]$
$\hat{\delta} = \frac{1}{2} \cdot \frac{y_{-1} - y_{+1}}{y_{-1} - 2y_0 + y_{+1}}$
$\hat{\tau}^{(1)} = k_{\mathrm{opt}} + \hat{\delta}$
Guard $D < -10^{-15}$; clip $\hat{\delta}$ to $[-0.5, 0.5]$
Stage 2: Local Grid Refinement
$\hat{\tau}_l = \arg\max_{\tau \in \mathcal{T}} \left|\langle \mathbf{s}(\tau), \hat{\mathbf{x}}_l \rangle\right|^2$
$(\Delta_1 = 0.05 \rightarrow \Delta_2 = 0.005)$
Previous delay, endpoints
and boundary protection
M-step Amplitude
$\hat{\alpha}_l = \frac{\langle \mathbf{s}(\hat{\tau}_l), \hat{\mathbf{x}}_l \rangle}{E_s}$
Residual Update
$\mathbf{r} \leftarrow \hat{\mathbf{x}}_l - \hat{\alpha}_l \cdot \mathbf{s}(\hat{\tau}_l)$
next path
next iter
After last path: joint amplitude fit
$S = [s_{\hat{\tau}_1}, \ldots, s_{\hat{\tau}_L}]$
$\hat{\alpha} = S^{\dagger} y$, $r = y - S\hat{\alpha}$
SAGE Loop: for $i = 1 : I_{\max}$, for $l = 1 : L$ $I_{\max} = 15$, $L = 3$